

TIME OF OSSIFICATION OF THE BONES OF THE
HAND OF THE MALE AND FEMALE
AND
UNION OF EPIPHYSES WITH THE DIAPHYSES

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In the course of my research on the time of appearance of centers of ossification in the bones of the carpus, and on the union of the epiphyses of the metacarpals, phalanges, the distal extremity of the radius and that of the ulna, I have found marked differences in the time element between the male and the female. I suppose these differences are just what one should expect, knowing that the female is developed in many ways sooner than the male.

There are instances of normal early ossification in both sexes, and there are normal cases in which ossification is delayed. In either instance I believe this is due in the main to heredity. I have followed a few of these cases of what may be called precocious and delayed ossification, and I think they are family characteristics.

Differences such as those here to be dealt with are found already in the embryo, the foetus and the new born child, and so on to the full development. In other articles published I have given the result of the study of over seven hundred roentgenograms, giving the time of appearance of centers of ossification and the chronological order of ossification. In the embryo the time of appearance of many of the centers of ossification has been well established by other investigators, but the difference in the sexes was not noted.

The studies here reported are based on carefully made roentgenograms of 64 white males, of 12 y. 6 mo. to 22 y. 10 mo. of age and 81 white females, 12 y. to 22 y. 6 mo. of age. In all the plates I have recorded the date of birth and the date when the plate was made, thus giving the exact age in years, months and days.

I have been asked, why the days, weeks, and later even the months? A very little study of the subject will show why the days in the embryo. Mall states the first point of ossification appears in the clavicle in the sixth week; there is then a rapid development, following in a few days of other points, and in a few weeks a large number have made their appearance. Such conditions follow throughout the intrauterine development, so that this is measured at first by days, then weeks, then

months. Then after birth why the weeks and months? The capitatum rarely shows a center of ossification before birth but the time of its appearance is measured by weeks and months. In the female it will have appeared in practically all cases when the child is six months of age and in the male by the tenth month. We may note in a film or plate an absence of beginning ossification of any of the bones, yet if a film is made a few days later it may show clearly beginning ossification. This in my opinion is of no small importance. When we consider the presence and the union of epiphyses from a surgical viewpoint, no one knows better than an experienced roentgenologist how important it is to know with reasonable accuracy the time of appearance of the epiphyses and their union with the shaft of the bone.

The first evidence of ossification of any of the epiphyses is seen as a pin point of bone tissue. This gradually enlarges and approaches the form of the bone. In the phalanges the ossifying "centers" soon become disc shaped and persist as such until fusion with the diaphysis, finally showing a faint line at the situation of union. When the union is complete the texture of the bone shows no further sign of demarcation.

The ossifying epiphyses of the metacarpal bones soon become round and persist in this form until they assume the shape of the end of the bone. The line of union of these epiphyses with the diaphyses is different from that in the phalanges. This line of union is always irregular and shows a zigzag appearance, until the texture of the bone can be seen, in complete union. This does not apply to the metacarpal bone of the thumb in which conditions are somewhat like those of the phalanges.

The individual data obtained in the present studies and their summary, are given briefly in the following tables:

No.	Age y m d	Sex	Distal end of Radius Ulna	UNION OF EPIPHYSES		Phalanges	
				Metacarpals			
1.	12-6-0	M	Distinct Gap ¹	Distinct Gap		Distinct Gap	
2.	12-9-4	"	" "	" "		" "	
3.	12-10-0	"	" "	" "		" "	
4.	12-10-21	"	" "	" "		" "	
5.	13-3-6	"	" "	" "		" "	
6.	13-7-25	"	" "	" "		" "	
7.	13-9-5	"	" "	" "		" "	
8.	14-10-6	"	" "	" "		" "	
9.	14-6-16	"	" "	" "		" "	
10.	15-2-19	"	" "	" "		" "	
11.	15-6-28	"	" "	" "		" "	
12.	15-11-25	"	" "	Distinct gap in all		D. G. in all	
13.	16-4-18	"	" "	Indistinct gap in all		I. G. in all	
14.	16-4-11	"	" "	Distinct " " "		Distinct gap in all	
15.	16-10-25	"	" "	Distinct " " "		D. G. in all but 2d 3d row line	

16.	16-10-25	"	D. gap	I. gap	I. G. in all	I. G. in all
17.	17-2-10	"	I. gap	"	Line in all	Line in all
18.	17-4-20	"	United	United	United in all	United in all 3d row line
19.	17-5-28	"	D. gap	D. gap	"	"
20.	17-5-0	"	"	"	D. G. in all	D. G. in all but 3d row line
21.	17-6-13	"	"	"	"	D. G. in all
22.	17-7-14	"	"	"	"	"
23.	17-7-16	"	I. G.	I. G.	United in all	United in all
24.	17-8-1	"	Distinct gap	"	Distinct gap in all	D. G. in all
25.	17-8-7	"	Indistinct gap	"	United in all	United in all
26.	17-9-15	"	D. G.	I. G.	Line in all	Line in all
27.	17-1-0	"	D. G.	D. G.	"	"
28.	17-9-0	"	I. G.	Line	United in all	United in all
29.	17-6-15	"	D. G.	I. G.	"	"
30.	17-11-18	"	"	"	I. G.	"
31.	18-0-0	"	"	"	D. G.	I. G.
32.	18-0-9	"	"	"	United	United
33.	18-1-2	"	"	I. G.	"	" 3d row line
34.	18-1-23	"	"	"	"	"
35.	18-2-16	"	"	D. G.	D. G.	D. G. 3d row line
36.	18-2-21	"	"	"	United	I. G.
37.	18-3-23	"	"	"	"	United
38.	18-5-20	"	"	"	"	"
39.	18-5-14	"	"	I. G.	"	" but 3d row line
40.	18-6-11	"	I. G.	"	"	"
41.	18-7-0	"	"	United	Line	Line
42.	18-10-10	"	D. G.	I. G.	I. G.	I. G.
44.	19-1-10	"	"	United	"	"
45.	19-3-1	"	"	I. G.	"	"
46.	19-4-20	"	I. G.	United	"	United " but 3d row line
47.	19-5-17	"	"	"	"	"
48.	19-6-21	"	"	"	Line	"
49.	19-11-5	"	United	"	United	"
50.	19-11-18	"	"	"	"	" 3d row line
51.	20-0-19	"	Line	"	"	"
52.	20-1-15	"	United	"	"	"
53.	19-0-6	"	Line	"	"	"
54.	20-3-29	"	United	"	"	"
55.	20-5-22	"	"	"	"	"
56.	20-5-21	"	"	"	"	" 3d row line
57.	20-5-22	"	"	"	"	"
58.	21-1-9	"	"	"	"	"
59.	21-4-16	"	"	"	"	"
60.	22-5-18	"	"	"	"	"
61.	22-5-20	"	"	"	"	"
62.	22-6-0	"	"	"	"	"
63.	22-8-0	"	"	"	"	"
64.	22-10-0	"	"	"	"	"
1.	12-1-9	F	Distinct gap	"	Distinct gap	Distinct gap
2.	12-5-9	"	"	"	"	"
3.	12-7-9	"	"	"	"	"
4.	13-10-17	"	"	"	"	"
5.	13-11-10	"	"	"	"	"
6.	14-3-21	"	"	"	United in all	Line in all
7.	14-3-10	"	"	"	Distinct gap in all	Distinct gap in all
8.	14-4-6	"	"	"	"	"
9.	14-4-9	"	"	"	United in all	United in all
10.	14-5-16	"	"	"	"	"
11.	14-6-0	"	"	"	Faint line in all	Faint line in all
12.	14-6-23	"	"	"	United in all	United in all
13.	14-7-11	"	"	"	Indistinct gap in all	Line in all

UNION OF EPIPHYSES—*Continued.*

No.	Age	Sex	Distal end of	Metacarpals	Phalanges
	<i>y m d</i>		Radius Ulna		
14.	14-8-13	F	Distinct gap	Distinct gap in all	Distinct gap in all
15.	14-9-4	"	"	"	"
16.	14-10-15	"	"	Indistinct gap in all	Faint line in all
17.	14-10-0	"	D. G. I. G.	I. G. in all	Line in all
18.	14-10-8	"	D. G. D. G.	"	"
19.	14-10-15	"	"	D. G. " "	D. G. " "
20.	15-3-7	"	"	United in all	United in all
21.	15-6-6	"	"	"	"
22.	15-6-20	"	I. G. United	"	"
23.	15-6-21	"	D. G. D. G.	Distinct gap in all	I. G. in all but last row line
24.	15-8-21	"	"	Line in all	United in all
25.	15-7-21	"	"	United in all	"
26.	16-0-14	"	"	"	"
27.	16-1-9	"	"	"	"
28.	16-2-27	"	Line United	"	"
29.	16-3-3	"	"	"	"
30.	16-3-15	"	Line	"	"
31.	16-4-21	"	I. G. "	Line in all	Line in all
32.	16-6-24	"	D. G. I. G.	Indistinct gap in all	I. G. in all
33.	16-7-0	"	Line United	United in all	United in all
34.	16-9-21	"	"	"	"
35.	16-9-2	"	"	"	"
36.	16-9-0	"	"	"	"
37.	16-10-0	"	D. G. D. G.	"	Line
38.	17-0-0	"	Line United	"	United
39.	17-0-24	"	D. G. I. G.	"	"
40.	17-2-14	"	Line United	"	"
41.	17-2-12	"	"	"	"
42.	17-3-21	"	United	"	"
43.	17-4-2	"	"	"	"
44.	17-6-0	"	"	"	"
45.	17-8-22	"	Line	"	"
46.	17-8-11	"	United	"	"
47.	17-8-19	"	Line	"	"
48.	17-9-11	"	United	"	"
49.	17-10-26	"	"	"	"
50.	17-11-24	"	"	"	"
51.	17-6-17	"	"	"	"
52.	17-10-25	"	"	"	"
53.	17-11-0	"	"	"	"
54.	17-11-10	"	"	"	"
55.	18-1-14	"	"	"	"
56.	18-3-14	"	Line	"	"
57.	18-3-24	"	United	"	"
58.	18-3-13	"	"	"	"
59.	18-5-0	"	"	"	"
60.	18-7-0	"	Line	"	"
61.	18-8-0	"	"	"	"
62.	19-0-22	"	United	"	"
63.	19-0-16	"	Line Line	"	"
64.	19-0-10	"	United United	"	"
65.	19-4-3	"	"	"	"
66.	19-4-3	"	"	"	"
67.	19-5-0	"	Line	"	"
68.	19-6-65	"	"	"	"
69.	19-7-0	"	United	"	"
70.	19-7-13	"	"	"	"

71.	19-11-18	F	Line	United	United in all	United in all
72.	20-0-17	"	United	"	" " "	" " "
73.	20-1-22	"	"	"	" " "	" " "
74.	20-3-5	"	"	"	" " "	" " "
75.	20-6-13	"	"	"	" " "	" " "
76.	20-8-2	"	"	"	" " "	" " "
77.	20-10-10	"	"	"	" " "	" " "
78.	21-5-10	"	"	"	" " "	" " "
79.	22-2-0	"	"	"	" " "	" " "
80.	22-0-8	"	"	"	" " "	" " "
81.	22-6-2	"	"	"	" " "	" " "

¹Abbreviations: D. G.=distinct gap.

I. G.=indistinct gap.

With the male hand I gave special attention to the ages of sixteen to twenty inclusive, in which we had forty-five roentgenograms, and with the female hand to the ages of fourteen to seventeen in which we had forty-nine roentgenograms.

Analysis of the data shows a distinct *cartilaginous gap* between the epiphyses and diaphyses in the hands of both male and female under thirteen years of age. In a few instances there may be found an indistinct gap between the epiphyses and the shaft of the phalanges of the first row in the female hand, indicating that union is taking place in the phalanges before that of the metacarpal bones. However I am inclined to think that complete union will be found in the terminal phalanges after it takes place in the others.

The sequence of union is not the sequence of development.

After the age of fourteen and up to fifteen, we find in the female hand complete union of all of the epiphyses of the hand, with the exception of those of the distal extremity of the radius and ulna, in many instances. In others the gap has been narrowed to what we call an indistinct gap, and in others only a faint epiphyseal line is seen, while in the male hand there is a *distinct gap* in every instance.

In the female hand from the age of fifteen to sixteen the union is complete in many individuals and when not complete only a faint epiphyseal line is seen, while in the male hand a *distinct gap* is seen with a very little narrowing in a few cases.

From the sixteenth to the seventeenth year in the female the union is complete in all the epiphyses except those of the radius and ulna, and many of those of the ulna show complete union, and only a faint epiphyseal line is seen in the radius, while in the male a *distinct gap* is present in all but a few instances where we find an indistinct gap and no evidence of complete union.

In the female at the age of seventeen years and three months, the union is complete in all the epiphyses including those of the radius and

ulna. Only nine of the remaining thirty-nine in the series of sixty-one hands show a faint epiphyseal line at the radius, while in the male hand at the age of seventeen years and three months *none* of the epiphyses show complete union with the shaft. It is only in rare instance that complete union is shown by the nineteenth year in the male hand. At twenty years of age or soon after the union seems complete in the male hand. The earliest hand of the female too shows complete union of the metacarpal epiphyses and the phalangeal epiphyses is that of No. 6, 14-3-21. The earliest in the male hand is that of 18, 17-4-20. The slowest in development in the female hand is that of 39, 17-0-24, in which there is a distinct gap at the radial epiphysis and an indistinct gap at the ulnar epiphysis but the metacarpal and phalangeal epiphyses have united.

The slowest in the male hand is that of No. 48, 19-6-21.

A careful study of these roentgenograms, and I believe this would be the same if we had an unlimited number, will show a difference of three years in the sexes in a large majority of the cases and a difference of four years in others.

The chronological order in which the bones of the carpus are ossified does not seem to differ, materially, in the sexes.

CHRONOLOGICAL ORDER								
<i>Male</i>								
	Capitulum	Hamatum	Triquetrum	Lunatum	Navicular	Lesser Multangular	Greater Multangular	Pisiform
First in order.....	114	2						
Second.....	2	112						
Third.....			94	11				
Fourth.....			8	79	1	2	2	
Fifth.....				3	25	13	17	
Sixth.....				1	10	13	8	
Seventh.....					2	4	12	
Eighth.....								
<i>Female</i>								
First in order.....	111	2						
Second.....	2	111						
Third.....			107	2				
Fourth.....			2	79	2	7	5	
Fifth.....				4	43	21	11	
Sixth.....				1	13	34	22	
Seventh.....				6	15	12	29	
Eighth.....								1

I have compiled the preceding from a former published table in which I gave the size of the centers of ossification and the order in which they appeared. These include in the female the ages of three months

and eight days to eight years, and we have one hundred and thirteen cases. With the male hand we have one hundred and sixteen individuals, from seven months and twelve days to eight years.

Matters seem to be following the same order until we get to the greater and lesser Multangular. In the female hand the lesser Multangular has thirty-four in the sixth position to twenty-two for the greater Multangular, while in the male hand the lesser Multangular has thirteen to eight for the greater Multangular, but the latter has seventeen in the fifth position and the lesser Multangular has thirteen. You will note, however, that while the greater Multangular has seventeen in the fifth position, the Navicular has twenty-five in this position in the male hand and the female has forty-three.

I am inclined to think this frequency of finding the greater Multangular in the fifth position in the male hand is the reason we find in our standard text books of Anatomy the fifth position given to this bone. In the early editions of Gray this bone was given as the fourth to appear but later on this was changed to the fifth position. And it has remained in this position for many years.

It is just possible that the proper position of the greater Multangular in the male hand is the sixth, certainly not the fifth, and its proper position is undoubtedly the seventh in the female hand. If this difference of chronological order exists, then we have another difference in the ossification of the male and female skeleton. These marked differences in the ossification of the male and female skeleton are easily proven. Let any roentgenologist take a film of the hand of a brother and sister twins at five years of age or a brother at five and a sister at three years of age. The advanced ossification in the female hand is easily seen.

I select the age of five for the male and three for the female in brother and sister because the differences in ossification in the sexes are so plainly seen at this period. I have several sets of twins of the same sex and twins of the opposite sex. These differences apply not only to ossification of the bones of the carpus but to the epiphyses of the metacarpal bones, those of the phalanges and those of the radius and ulna. These marked differences continue until we have the union of the epiphyses with the diaphyses which takes place three to four years sooner in the female than in the male.

I have only investigated the question of unions of the epiphyses in connection with those of the hand, but have expressed an opinion as to the time of appearance of the epiphyses at the distal extremity of the femur, and the proximal extremity of the tibia.

In a few instances in which I found an unusual early union of the epiphyses of the hand, I have made plates of the elbow, knee and ankle. In all of these I found a corresponding early union of these epiphyses. The bones of the tarsus, the metatarsus and the phalanges of the foot ossify earlier than those of the hand.

We frequently find large centers of ossification in the calcaneum and talus before birth. A center of ossification in the cuboid is frequently found before birth and there is the same difference in the male and female skeleton that we find in the hand.

To quote from Stevenson, "The extent to which the venerable hand of tradition still controls our teaching on the subject of the time element

TIME OF APPEARANCE OF CENTERS OF OSSIFICATION IN THE EPIPHYSES OF THE HAND

<i>Metacarpal</i>	First.....		F Early in third year M Early in third year	
	Second.....		F Late in first year M Early in second year	
	Third.....		F Early in second year M Late in second year	
	Fourth.....		F Early in second year M Late in second year	
	Fifth.....		F Early in second year M Late in second year	
		1st row	F Late in first year M Early in second year	
	Middle	2d row	F Early in second year M Late in second year	
		3d row	F Late in second year M Late in third year	
		1st row	F Early in second year M Late in second year	
	Ring	2d row	F Early in second year M Early in third year	
3d row		F Late in second year M Late in third year		
1st row		F Early in second year M Late in second year		
<i>Phalanges</i>	Index	2d row	F Early in second year M Early in third year	
		3d row	F Late in second year M Late in third year	
		1st row	F Early in second year M Late in second year	
	Little	2d row	F Late in second year M Early in third year	
		3d row	F Late in second year M Early in third year	
		1st row	F Late in second year M Early in third year	
	Thumb	1st row	F Late in second year M Early in third year	
		2d row	F Early in second year M Late in second year	
	<i>Radius</i>			F About eighth month M About 15th month
	<i>Ulna</i>			F Bet. 6th and 7th yr. M Bet. 7th and 8th yr.

in epiphyseal union is seen in the frequency with which expressions found in the earliest editions of well known texts are retained practically verbatim in some of the latest editions of these same works."

Why should our standard text books of Anatomy continue the statement year after year that the epiphyses of the metacarpal bones and the phalanges of the hand appear "about three years" or "third to fifth year," when many of these points appear in the female hand before the child is one year of age, others before two years and in many instances all have appeared before three years of age.

In the male hand a large number have appeared before the child is three years of age. The table on Epiphyses is the result of the study of sixty-four female and sixty-three male hands under six years of age.

I must, however, take issue with the statement made by Stevenson that the findings in the study of radiograms are not reliable. It is my opinion that the X-ray is the most reliable method we have.

In good roentgenograms the texture of the bone is shown and the very faintest epiphyseal line may be seen showing an incomplete union. I do not believe this could be as plainly seen by the naked eye in the dried subject. The exact age in years, months and days can be obtained in the living subject. I regret that the author saw fit to state that I made the assertion "that epiphyses in the female unite three to four years earlier than in the male without any confirmatory evidence." Of course he did not see my publication of 1916.

I made the statement to which he refers, after the investigation of the one hundred and forty-five roentgenograms given in Table 1.

I think the hand is a very good index to the entire skeleton. I do not mean by this that all of the epiphyses are united when we find those of the hand united, but when this takes place early in the hand there is a corresponding early union of the other epiphyses and the marked difference between the male and female are seen to be present.

In many instances the hand of the female is found to be four years in advance of the male, and in the majority of instances the female is three years in advance of the male when completely developed. This applies not only to the appearance of points of ossification but to the union of epiphyses of the hand and the natural conclusion is it applies to all of the epiphyses of the skeleton.

SUMMARY AND CONCLUSIONS

The ossification of the bones of the female is in advance of the male from time of appearance of centers of ossification in the Embryo. The difference is more apparent as the Fetus develops. After birth the differ-

ences are still more marked until the union of the epiphyses with the diaphyses.

The appearance of centers of ossification in the epiphyses of the metacarpals and phalanges in the female hand are decidedly earlier than in the male hand. A few have made their appearance in the female before the close of the first year and nearly all have appeared before the close of the second year; while in the male hand the most of them appear late in the second year or early in the third year.

The order of ossification of the bones of the carpus is apparently the same in the sexes.

The examination of 81 roentgenograms of the hands of females from 12 to 22 years of age, and 64 roentgenograms of hands of males from 12 to 22 years of age show that the union of the epiphyses with the diaphyses takes place from three to four years sooner in the female than in the male.

The hand is a good index of the extent of ossification in the entire skeleton, an early union of the epiphyses of the hand indicates a corresponding early union of all of the epiphyses.

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